Skills Summary

Integration by Parts: Choosing u and dv Strategically

Use this reference while completing your worksheet or when studying.

The formula is one line: $\int u\,dv = uv - \int v\,du$. All of the skill is in the choice. A good choice makes $\int v\,du$ *simpler* than what you started with; a bad one makes it worse, and the fastest fix is to swap u and dv and start again.

1. Choosing u and dv: one application

Formula: $\displaystyle\int u\,dv = uv - \int v\,du.$ **Tie-breaker (LIATE):** take u from the earliest category present — **L**og, **I**nverse trig, **A**lgebraic, **T**rig, **E**xponential — and dv from the latest.

The reason behind the letters: u should get *simpler* when differentiated, and dv must be something you can actually integrate.

Example: Find $\displaystyle\int xe^{3x}\,dx$.

Step 1. A power times an exponential: differentiating x removes it, and e^{3x} integrates easily, so $u = x$ and $dv = e^{3x}dx$.
Step 2. $du = dx$ and $v = \frac{1}{3}e^{3x}$, so $\int xe^{3x}dx = \frac{x}{3}e^{3x} - \frac{1}{3}\int e^{3x}dx$.

$$\boldsymbol{\frac{x}{3}e^{3x} - \frac{1}{9}e^{3x} + C}$$

Try it: Find $\displaystyle\int xe^{-x}\,dx$.

check: $\boldsymbol{-xe^{-x} - e^{-x} + C}$

2. When a logarithm is present, it is always u

Rule: $\ln x$ has no elementary antiderivative you can use as v, so it can only be u — and that is lucky, because $\frac{d}{dx}\ln x = \frac{1}{x}$ turns it into a power.

If the logarithm stands alone, supply the missing factor: $\ln x = \ln x \cdot 1$, so $dv = dx$ and $v = x$.

Example: Find $\displaystyle\int x^3 \ln x\,dx$.

Step 1. Taking $u = x^3$ would need $v = \int \ln x\,dx$, another parts problem; taking $u = \ln x$ replaces it by $\frac{1}{x}$, so choose that.
Step 2. $v = \frac{x^4}{4}$, and $\int \frac{x^4}{4} \cdot \frac{1}{x}dx = \int \frac{x^3}{4}dx = \frac{x^4}{16}$.

$$\boldsymbol{\frac{x^4}{4}\ln x - \frac{x^4}{16} + C}$$

Try it: Find $\displaystyle\int x^2 \ln x \, dx$.

check: $\boldsymbol{\frac{x^3}{3} \ln x - \frac{x^3}{9} + C}$

3. When parts has to run twice

Two ways it repeats. If u is a power, each round lowers the degree by one, so x^2 needs two rounds and x^3 needs three — keep the same style of choice each time.

If neither factor simplifies (exponential times sine or cosine), nothing ever terminates: parts twice returns the original integral. Name it I, apply parts twice, then solve the resulting equation for I algebraically.

Example: Find $\displaystyle\int e^x \cos x \, dx$.

Step 1. Neither e^x nor $\cos x$ gets simpler, so aim for the cycle rather than for termination: set I equal to the integral.
Step 2. Parts twice (with u the trig factor both times) gives $I = e^x \cos x + e^x \sin x - I$, so $2I = e^x(\sin x + \cos x)$.

$$\boldsymbol{\frac{e^x(\sin x + \cos x)}{2} + C}$$

Try it: Find $\displaystyle\int e^{2x} \sin x \, dx$.

check: $\boldsymbol{\frac{e^{2x}(2\sin x - \cos x)}{5} + C}$

4. Parts in definite and improper integrals

Definite: $\displaystyle\int_a^b u \, dv = \Big[uv\Big]_a^b - \int_a^b v \, du$ — the boundary term is evaluated, the leftover integral keeps the same limits. No $+C$.

Improper: replace the infinite limit by t, integrate, and take $\lim_{t\to\infty}$ at the end. Never substitute infinity into an antiderivative. Useful fact: $t^n e^{-t} \to 0$ and $\frac{\ln t}{t} \to 0$, because polynomials beat logarithms and exponentials beat polynomials.

Example: Find $\displaystyle\int_0^\infty x e^{-x} \, dx$.

Step 1. The upper limit is infinite, so integrate to a finite t first; parts applies exactly as usual with $u = x$, $dv = e^{-x} dx$.
Step 2. $\int_0^t x e^{-x} dx = [-xe^{-x} - e^{-x}]_0^t = -te^{-t} - e^{-t} + 1$, and both t terms tend to 0.

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Try it: Find $\displaystyle\int_0^\infty x e^{-2x} \, dx$.

check: $\boldsymbol{\frac{1}{4}}$